

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2025

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. Let G be a group and let N be a normal subgroup. Let $C = \{g \in G \mid gng^{-1} = n \text{ for all } n \in N\}$.
 - (a) Show that C is a normal subgroup of G .
 - (b) Show that if N is finite then C has finite index in G . (*Hint:* $\text{Aut}(N)$ is finite.)
2. (a) Suppose A and B are $n \times n$ matrices with complex entries and suppose that $AB = BA$. Show that if A is diagonalizable with distinct eigenvalues, then B is diagonalizable.
 - (b) Let A be an $n \times n$ matrix with complex entries and suppose A has n distinct eigenvalues. Show that the $n \times n$ matrices that commute with A form a vector space of dimension n .
 - (c) Let A be an $n \times n$ matrix with complex entries and suppose A is diagonalizable but has at least one repeated eigenvalue. Show that the $n \times n$ matrices that commute with A form a vector space of dimension strictly larger than n .
3. Let R be a commutative ring with 1 and let J be the intersection of all the maximal ideals of R .
 - (a) Show that if $x \in J$ then $1 + x$ is a unit of R .
 - (b) Let I be an ideal of R . Suppose that $1 + y$ is a unit for each $y \in I$. Show that $I \subseteq J$.

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4. (a) Let R be a principal ideal domain and let M be a finitely generated R -module. Show that M is projective if and only if $\text{Hom}_R(M, R) \simeq \text{Hom}_R(R, M)$. (The isomorphism is an isomorphism of R -modules, where $r\phi$ is defined by $(r\phi)(x) = r(\phi(x))$ for a homomorphism ϕ .)
- (b) Give an example of a commutative ring R with 1 and a projective R -module M such that $\text{Hom}_R(M, R)$ and $\text{Hom}_R(R, M)$ are not isomorphic as R -modules. (*Note:* R could be a field and M could be infinitely generated.)
5. Suppose $L/\mathbb{Q}$ is a Galois extension such that $\text{Gal}(L/\mathbb{Q}) = A_5$, the group of even permutations of 5 letters. Show that there is an irreducible polynomial $f(x) \in \mathbb{Q}[x]$ of degree 20 such that L is the splitting field of $f(x)$. (You may use without proof the fact that A_5 is a simple group of order 60.)
6. The following is the character table for a group G .

	I	II	III	IV	V
#	1	15	20	*	12
ρ_1	1	1	1	1	1
ρ_2	3	-1	0	$(1 + \sqrt{5})/2$	$(1 - \sqrt{5})/2$
ρ_3	3	-1	0	$(1 - \sqrt{5})/2$	$(1 + \sqrt{5})/2$
ρ_4	4	0	1	**	-1
ρ_5	5	1	-1	0	0

The first line gives names to the conjugacy classes of G , and the second line gives the sizes of the conjugacy classes.

- (a) What is the order of G ?
- (b) What is the missing number denoted by * ?
- (c) What is the missing number denoted by ** ?
- (d) Show that G has no proper normal subgroup H such that G/H is abelian.